Data Card: Early-Warning Burnout Detection System

Stress Detection in Social Media Communications

Last Updated: August 2024
Dataset: Dreaddit (Reddit Stress Analysis) + Kaggle Stress Dataset

1. Dataset Overview & Purpose

Problem Statement

Students, professionals, and everyday individuals silently enduring burnout often slip through the cracks of digital support systems because their distress is masked by subtle, indirect language about daily commitments rather than explicit cries for help. This project addresses the critical need for early detection of stress and burnout through AI-assisted screening, enabling counsellors and community moderators to offer timely, proactive outreach.

Dataset Purpose

This data card documents a stress detection dataset designed to train machine learning models that can identify nuanced linguistic markers of psychological stress, anxiety, burnout, and emotional distress in social media communications. The dataset serves as a foundation for building privacy-respecting, culturally-sensitive early-warning systems that empower human caregivers to intervene before silent suffering escalates into severe mental health crises.

2. Input vs. Output

Dimension	Description
Input Data Type	Text-based social media posts from Reddit communities (subreddits focused on mental health, financial stress, interpersonal conflict, PTSD, and anxiety). Posts are typically 100-600 tokens in length and represent first-person narratives of personal experiences, help-seeking behavior, and expressions of distress.

Input Features	• Raw text content (post body) • Metadata: posting time, upvote ratio, karma score, comment count • Linguistic features: LIWC word categories (negative emotions, anxiety, social words), syntactic complexity measures (Flesch-Kincaid Grade, ARI), sentiment scores
Output Format	Binary classification with confidence scores: "Stress" (1) or "Not Stress" (0). Optional: segment-level stress localization (identifying which parts of a post express stress).
Output Use Case	Flagging high-confidence stress indicators for human review by counsellors or moderators; enabling proactive outreach; informing community support prioritization; generating case summaries for mental health professionals.

3. Classification Task: Labels & Label Definitions

Labels

Label	Definition	Frequency
Stress (1)	The author expresses mental or emotional strain, tension, or distress resulting from adverse or demanding circumstances. The text demonstrates both stress indicators AND a negative attitude about the situation. Examples: explicit mentions of anxiety, fear, dread, helplessness; description of physical stress symptoms (nausea, insomnia); first-person accounts of overwhelming situations.	52.3% (1,857/3,553 labeled segments)
Not Stress (0)	The author does not express psychological distress, OR expresses a stressful situation but frames it positively/retrospectively without negative emotional content. Examples:	47.7% (1,696/3,553 labeled segments)

	retrospective storytelling without current distress; positive reframing of challenges; social support narratives.	

Label Selection Rationale

Binary classification was chosen because: (1) it enables clear, actionable screening decisions for frontline support workers; (2) it aligns with clinical mental health assessment frameworks that distinguish distress from non-distress states; (3) it provides interpretability—humans can understand why a post was flagged; (4) binary outputs reduce alert fatigue for community moderators while maintaining sufficient sensitivity for early intervention.

4. Labeling Strategy

Data Annotation Process

The Dreaddit dataset used as a foundation employed the following annotation methodology:

- • Annotators: English-speaking Amazon Mechanical Turk workers (minimum 5 annotations per segment)
  • Unit of Analysis: Five-sentence text segments (~100 tokens) from longer Reddit posts
  • Label Options: "Stress," "Not Stress," or "Can't Tell" (segments with >50% "Can't Tell" votes were excluded)
  • Quality Control: 50 pre-verified check questions embedded in each batch; strict annotation guidelines provided
  • Inter-Rater Reliability: Fleiss's Kappa $\kappa = 0.47$ ("moderate agreement"); 39% perfect agreement, 32% majority $\geq 3/5$ votes

Label Quality & Consistency

Quality measures include: (1) majority voting from multiple independent annotators; (2) exclusion of low-confidence annotations; (3) stratification by agreement level for model training (separate experiments on 80%, 100% agreement thresholds); (4) explicit annotation instructions defining stress as both a condition AND a negative emotional response; (5) domain expertise review of guidelines by Columbia University researchers (Turcan & McKeown, 2019).

Ensuring Consistency in New Labeling

For any future annotation cycles on new data, we recommend: (1) reusing the validated Dreaddit annotation guidelines; (2) requiring >70% inter-rater agreement on a pilot subset before full-scale annotation; (3) implementing automated quality checks (e.g., agreement analysis, anomaly detection in labeling patterns); (4) maintaining a labeled reference set for

annotator calibration; (5) involving domain experts (mental health professionals, lived-experience advocates) in periodic guideline refinement.

5. Data Creation & Sources

Primary Dataset: Dreaddit (Reddit)

Attribute	Details
Collection Method	Scraping via PRAW API from Reddit public subreddits (January 2017 – November 2018)
Total Posts	187,444 posts across 10 subreddits
Labeled Segments	3,553 five-sentence segments labeled by Mechanical Turk annotators
Domains/Communities	abuse (r/domesticviolence, r/survivorsofabuse), anxiety (r/anxiety, r/stress), financial (r/almosthomeless, r/assistance, r/food_pantry, r/homeless), PTSD (r/ptsd), social/interpersonal (r/relationships)
Geographic Scope	Primarily English-speaking Reddit users; North American predominance
Temporal Coverage	2017–2018 (historical data); platform dynamics may differ in 2024
Average Post Length	420 tokens (much longer than Twitter; enables nuanced analysis)
Data Quality	Raw Reddit posts as-posted (includes spelling errors, informal language; representative of real social media)

Secondary Dataset: Kaggle Stress Analysis Dataset

Reference:
https://www.kaggle.com/datasets/payaldhokane/stress-analysis-in-social-media-dataset

This supplementary dataset provides additional stress-labeled social media communications for model validation and cross-domain testing. It enables assessment of model generalization across different data sources and annotation methodologies.

Data Composition by Domain

Domain	Subreddits	Total Posts	Labeled Segments
Abuse/Interpersonal Conflict	r/domesticviolence, r/survivorsofabuse	2,901	703
Anxiety & Mental Health	r/anxiety, r/stress	59,208	728

Financial Hardship	r/almosthomeless, r/assistance, r/food_pantry, r/homeless	12,517	717
PTSD	r/ptsd	4,910	711
Relationships & Social Stress	r/relationships	107,908	694

6. Web Collection Ethics & Legal Considerations

Consent & Privacy

✓ Reddit Data: Posts are from public subreddits with terms of service permitting academic research and data scraping (PRAW API is officially supported). However, important caveats apply:

• Individual user consent: Reddit users post publicly but may not explicitly consent to inclusion in ML datasets. Our approach prioritizes minimizing user identification and de-identification.
• De-identification: The original Dreaddit dataset removes usernames and identifiable metadata; we follow this standard.
• Subreddit selection: We focus on communities explicitly oriented toward help-seeking (mental health, financial assistance), where sharing struggles is the intended use of the platform.
• User notification: In deployment, platforms should notify users that their communications may be analyzed for mental health screening.

Copyright & Attribution

✓ Academic Use: The Dreaddit dataset is released for academic research under appropriate licensing (available at: http://www.cs.columbia.edu/~eturcan/data/dreaddit.zip).

✓ Attribution: We cite Turcan & McKeown (2019) and acknowledge Reddit as the data source.

✓ Commercial Use: Commercial deployment requires explicit license negotiation with Reddit and potential user notification/consent mechanisms.

Legal Compliance

✓ GDPR & Regional Privacy Laws: For deployment in EU/other regulated regions:
• Conduct Data Protection Impact Assessments (DPIA)
• Implement anonymization/pseudonymization where required
• Establish legal bases for processing (e.g., legitimate interest in mental health intervention,

with explicit consent for sensitive categories)

✓ Platform Terms of Service: Ensure compliance with Reddit's terms (Section 5: User Content) and any applicable research agreements.

✓ Health Data Regulations: If integrated with healthcare systems, comply with HIPAA (US), NHS Digital guidelines (UK), or equivalent standards.

Ethical Safeguards in Design

1. 1. Minimal data retention: Store only data necessary for model inference; implement data retention policies.
   2. Human oversight: All automated stress flags are reviewed by trained human moderators before any outreach.
   3. Opt-out mechanisms: Users should have clear ability to opt out of AI screening where applicable.
   4. Transparency: Communicate to users/platforms how data will be used.
   5. No secondary use: Do not repurpose stress data for advertising, discrimination, or surveillance.

7. Bias & Representation Analysis

Known Limitations & Potential Biases

Bias Category	Description	Mitigation Strategy
Language	Dataset contains only English-language posts. Stress expression varies significantly across languages and cultures.	Plan multilingual dataset expansion (Spanish, Mandarin, Arabic, French, etc.); validate with native speakers.
Geographic/Cultural	Reddit predominately North American; stress expressions may not generalize to other cultures (e.g., collectivist vs. individualist stress narratives).	Partner with international research teams; incorporate posts from region-specific communities; conduct cross-cultural validation.
Socioeconomic	Certain domains over-represented (e.g., r/relationships dominant with 107K posts). Financial stress posts fewer (12.5K). May miss stress expressions from lower-internet-access populations.	Balance domain sampling; explore emerging platforms (e.g., WhatsApp, local forums) for underrepresented populations; engage community organizations.

Mental Health Framing	Dataset reflects medicalized/Western psychology definitions of stress. May miss indigenous or alternative frameworks of distress.	Consult with diverse mental health practitioners; incorporate peer support and lived-experience perspectives in labeling.
Platform Bias	Reddit users skew younger, more educated, higher tech literacy. Older adults, less digitally-fluent populations underrepresented.	Incorporate data from age-diverse platforms (Facebook communities, community forums); test generalization across age groups.
Selection Bias	Only captures individuals who seek help online; misses silent sufferers with no digital presence.	Acknowledge limitation in system design; supplement with offline data collection (surveys, clinical data, community outreach) where feasible.
Annotation Bias	Moderate inter-rater agreement (κ=0.47) suggests subjectivity in stress labeling. Annotators may have implicit biases regarding certain stress expressions.	Use diverse annotator pools; conduct bias audits on annotation disagreement patterns; involve mental health experts in guideline refinement.

Improving Diversity & Representation

1. Multilingual & Cross-Cultural Expansion:
   • Partner with international research institutions to collect and annotate stress data in underrepresented languages.
   • Validate cultural appropriateness of stress definitions with indigenous communities and mental health experts.

2. Intersectional Sampling:
   • Actively seek posts from LGBTQ+, disability, racial/ethnic minority communities; audit for representation.
   • Stratify sampling to ensure balanced representation across age, gender, and socioeconomic markers.

3. Lived-Experience Integration:
   • Include individuals with lived experience of burnout and mental health crises in dataset design and annotation.
   • Conduct participatory research with affected communities (co-design approach).

4. Bias Auditing:
   • Conduct regular fairness evaluations by demographic group (age, gender, race, disability

status where applicable).
  • Report false positive/false negative rates stratified by demographic; mitigate disparities.

5. Offline Data Integration:
  • Supplement social media data with clinical records (with appropriate consent), community health surveys, and focus group insights.
  • Test model performance across data sources and populations.

8. Values & Community Alignment

Core Values Embedded in Dataset Design

Value	Operationalization in Dataset	Community Benefit
Privacy & Dignity	De-identified data; no collection of user IDs, profile information, or social networks. Minimal data retention policies. User opt-out mechanisms in deployment.	Individuals' identities are protected; data cannot be weaponized for surveillance or discrimination.
Human Agency	Humans (counsellors, moderators) retain control over all interventions. ML provides flagging/screening only; no autonomous decisions about care.	Support workers are empowered, not replaced. Lived-experience perspectives guide intervention design.
Equity & Inclusion	Deliberate over-sampling of marginalized communities (abuse survivors, people with PTSD, low-income individuals). Linguistic and cultural diversity prioritized.	Silent suffering in minority communities is visible and prioritized for support. Reduces care disparities.
Transparency	Clear documentation of dataset composition, annotation methodology, model limitations, and deployment guidelines. Open-source code and reproducible research.	Communities understand how their data is used. Enables external audit and accountability.
Harm Prevention	Designed to enable early intervention (reduce crisis escalation), not punishment or control. Strict ethical guidelines on data use.	System supports recovery, not coercion. Reduces stigma by positioning stress detection as universal health screening.

Accountability	Dataset documentation includes bias audits, fairness reports, and community feedback loops. Regular re-evaluation with stakeholders.	Project team remains accountable to communities served. Continuous improvement based on lived experience.

Alignment with Challenge Problem Statement

This dataset directly operationalizes the Challenge problem statement:

✓ "Students, professionals, and everyday individuals silently enduring burnout..."
→ Dataset captures diverse domains (anxiety, PTSD, financial hardship, relationship stress, abuse) reflecting varied burnout contexts.

✓ "...masked by subtle, indirect language..."
→ Average post length (420 tokens) vs. microblogs enables detection of nuanced, non-explicit stress markers. LIWC and syntactic analysis captures linguistic subtlety.

✓ "...AI solution acts as an early-warning screening tool..."
→ Binary stress/not-stress classification with confidence scoring enables prioritization for human review.

✓ "...strict privacy, safety, and inclusivity..."
→ De-identified data, human-in-the-loop design, bias mitigation strategies, and community consultation ensure ethical deployment.

✓ "...bridge the gap between silent suffering and immediate care..."
→ By identifying linguistic patterns of distress, the system enables proactive outreach by counsellors and moderators before crises escalate.

Community Accountability & Feedback

To remain accountable to communities served, we commit to:

1. Community Steering Committee: Quarterly meetings with mental health professionals, peer support workers, and individuals with lived experience to review dataset performance and ethical impact.

2. Transparency Reports: Annual publication of fairness audits, demographic performance breakdowns, and documented model failures.

3. Feedback Mechanisms: Clear pathways for Reddit communities and users to report concerns about data use or model errors.

4. Benefit-Sharing: Commit that any commercial deployment will fund mental health

initiatives, research, or support services in underserved communities.

5. Data Governance: Establish clear protocols for data deletion, retention, and future use. No secondary applications without explicit community consent.

9. Team & Collaborators

This Data Card was developed collaboratively. Please add your team members below:

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10. References

- Turcan, E., & McKeown, K. (2019). "Dreaddit: A Reddit Dataset for Stress Analysis in Social Media." arXiv preprint arXiv:1911.00133.
- Kaggle Dataset: Dhokhane, P. "Stress Analysis in Social Media Dataset." https://www.kaggle.com/datasets/payaldhokane/stress-analysis-in-social-media-dataset
- Data Cards for Dataset Documentation (Parkdale et al., 2022): https://pair-code.github.io/datacardsplaybook/
- Model Cards for Model Reporting (Mitchell et al., 2019): https://arxiv.org/abs/1810.03993

- AI Ethics & Responsible AI Frameworks:
- • Microsoft: Fairness, Accountability, Transparency (FAT) in AI
- • Google AI: Responsible AI Practices
  https://ai.google/responsibilities/responsible-ai-practices/
- • Partnership on AI: AI & Society frameworks
- Mental Health & Technology Ethics:
- • Mindstrong & Digital Mental Health Ethics (https://mindstrong.com)
- • WHO Guidelines on Digital Mental Health Interventions

Appendix: Example Labeled Segments

Example 1: PTSD Domain (Stress)

- "I only get it when I have a flashback or strong reaction to a trigger. I notice it sticks around even when I feel emotionally calm and can stick around for a long time after the trigger, like days or weeks. Its a new symptom I think. Also been having lots of nightmares again recently. Not sure what to do as Im not currently in therapy, but I am waiting to be seen at a mental health clinic."

Label: STRESS | Annotator Agreement: 6/7 (86%)

Stress Indicators: New symptom, nightmares, uncertainty, helplessness, help-seeking behavior

Example 2: Abuse Domain (Not Stress - Retrospective)

- "Regardless, that didn't last long, maybe half a year. I released that apartment, and most of my belongings (I kept a few boxes of my things from the military, personal effects, but little else). Looking back, there were some signs of emotional manipulation here, but it was subtle... and you know how it is, love is blind. We got engaged. It was quite the affair."

Label: NOT STRESS | Annotator Agreement: 5/5 (100%)

Reason: Retrospective framing with emotional distance; no current distress expressed despite traumatic context

Example 3: Financial Domain (Stress)

- "Our dog Jett has been diagnosed with diabetes and is now in the hospital to stabilize his blood sugar... his bill is large enough that we just can't cover it on our own (especially with our poor financial situation). We're being evicted from our home soon and trying to find a place with this bill is just too much for us by ourselves."

Label: STRESS | Annotator Agreement: 3/5 (60%)

Stress Indicators: Multiple compounding stressors (illness + financial hardship + homelessness risk); overwhelm; help-seeking